

EVOLUTION OF ARTIFICIAL INTELLIGENCE

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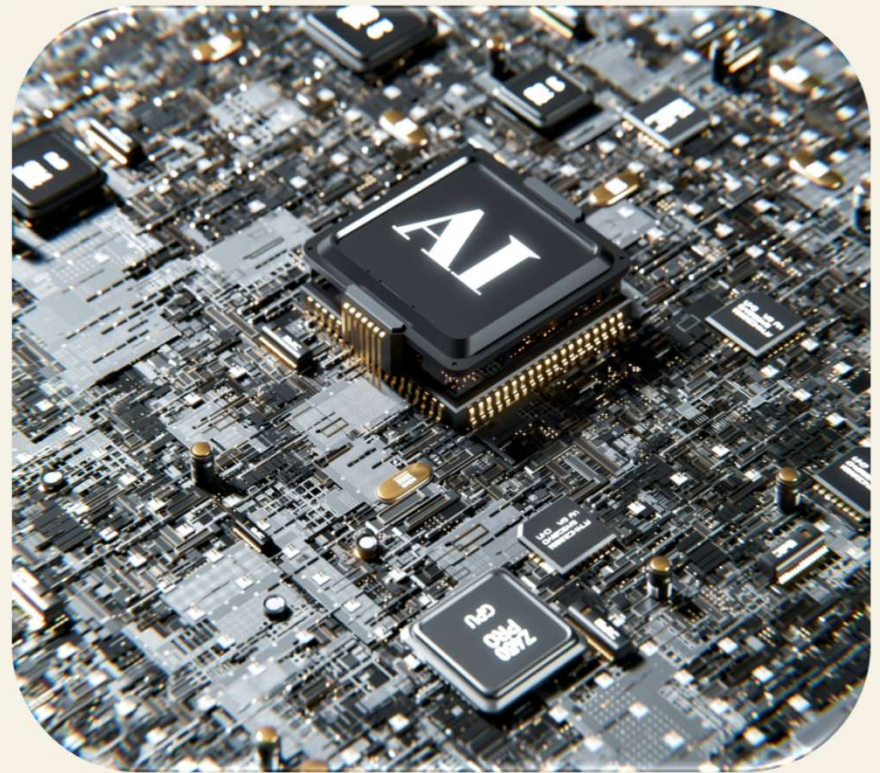
From Mid-20th Century Mathematical Foundations to the Generative AI Boom of 2026. A comprehensive chronological mapping of paradigms, innovations, and winter cycles

The Broad Context

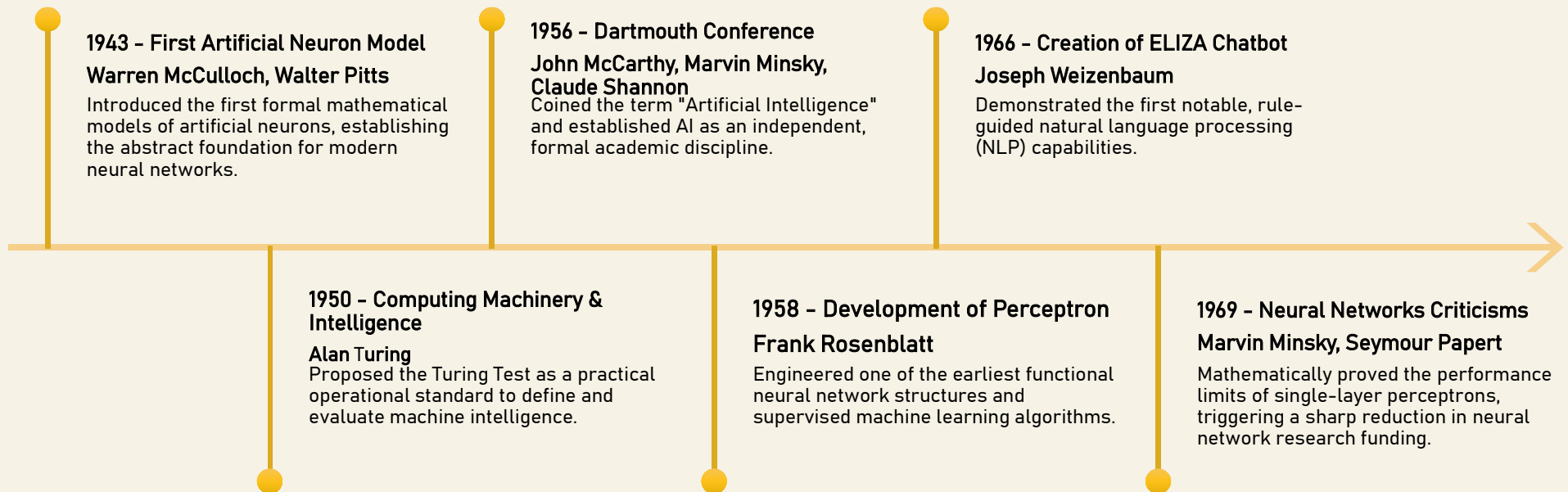


A Transformative Trajectory

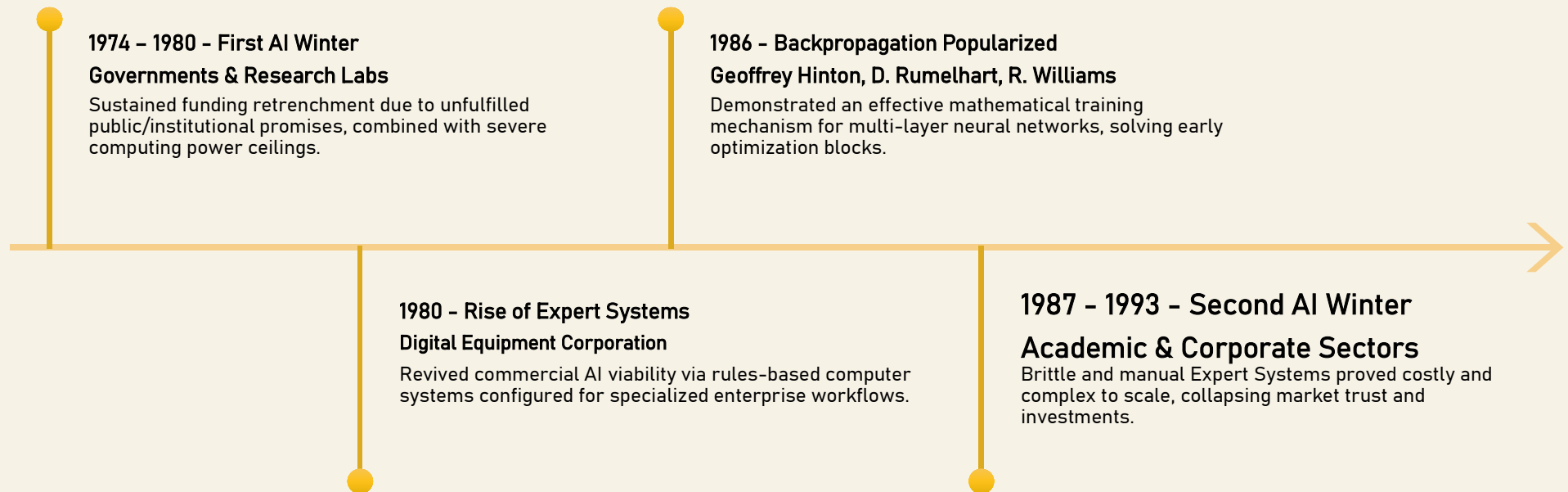
- Artificial Intelligence (AI) has evolved from theoretical concepts in the 1940s to a major driver of modern innovation.
- Its growth has been driven by advances in computational methods, hardware technology, and data availability.
- AI development has experienced cycles of rapid progress and periods of decline in interest and funding, known as "AI Winters."



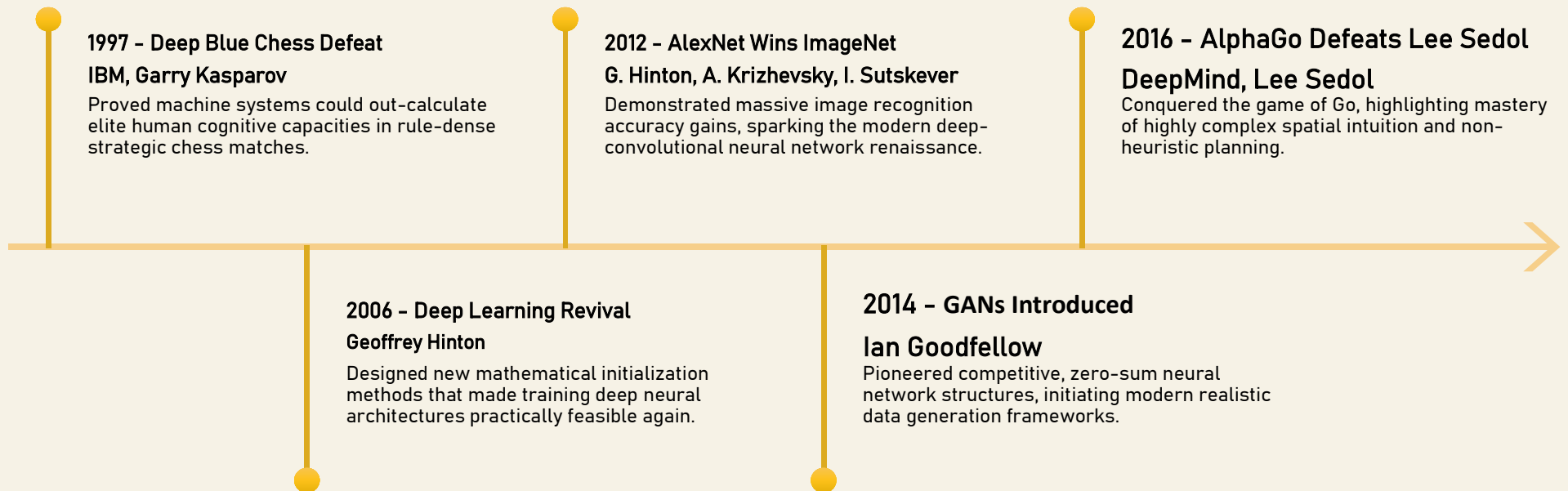
Chronological Era I: Foundations (1943 – 1969)



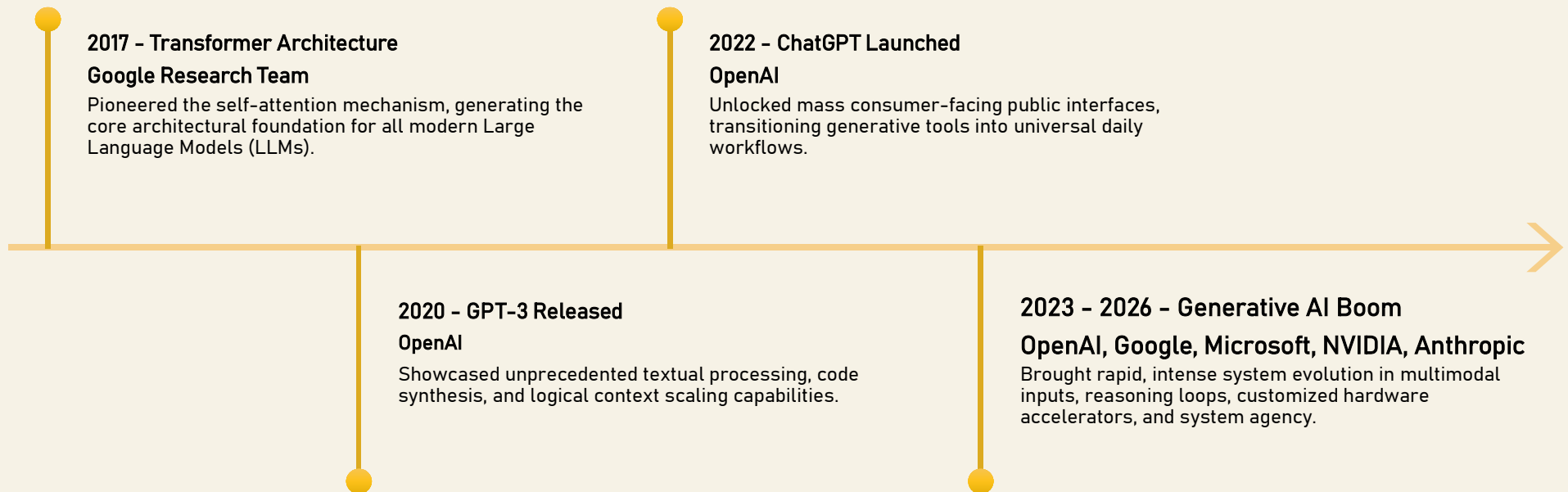
Chronological Era II: Winters & Resurgence (1974 - 1993)



Chronological Era III: Machine Superiority (1997 – 2016)



Chronological Era IV: Generative Era (2017 - 2026)



AI Winters Explained

First AI Winter (1974 - 1980)

The Ceiling of Computation

- Triggered as research groups and funding agencies experienced intense, unmet expectations regarding initial simple neural models.
- Processing hardware at the time lacked computational speed, memory capacity, and storage densities required to compute any substantial real-world datasets.

Second AI Winter (1987 - 1993)

The Collapse of Expert Systems

- Driven primarily by the commercial limits and massive operational fragility of handcrafted rule-based "expert systems."
- These systems proved too complex to scale manually, were highly expensive to design and maintain, and repeatedly failed under non-deterministic edge cases.

Modern Success Drivers



Three Technological Pillars

- **Deep Learning Networks:** Advanced neural networks enabled automatic learning from complex visual and textual data.
- **Specialized Hardware (GPUs):** Parallel-processing hardware significantly reduced AI model training times.
- **Scalable Cloud Computing:** Cloud infrastructure provided accessible, large-scale computing power and data storage, accelerating AI development globally.



Classification: Narrow vs. General AI

Narrow (Weak) AI

Highly Specialized Solutions

- AI systems engineered to excel at isolated tasks, such as translation pipelines, semantic search engines, or autonomous driving algorithms.
- All current operational AI architectures, including state-of-the-art LLMs, remain fundamentally classified under the Narrow AI category.



Artificial General AI (AGI)

Theoretical Universal Adaptation

- A theoretical intelligence model capable of learning, adjusting, and resolving any conceptual challenge equivalent to a human mind.
- While generative models continue to scale rapidly, the path and safety protocols required to reach AGI remain intensely debated.

Market Leaders & Models (2026)

Company	Flagship Models & Tech Systems	Strategic Focus
OpenAI	GPT-4o, ChatGPT, GPT-x series	Conversational engines & multimodal reasoning models
Google DeepMind	Gemini series, AlphaGo, AlphaFold	Scientific discovery & consumer multimodal ecosystem
Anthropic	Claude series (Claude 3 / 3.5 / 4)	Highly aligned, constitutionally secure assistant systems
Meta	Llama open models series	Democratic, open-source model infrastructures
Microsoft	Copilot platform integrations	Productivity stack and cloud scale API platforms
NVIDIA	AI Accelerators, H200/Blackwell Architecture	Hardware acceleration & physical infrastructure computing

Academic References



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